Yunsoo Kim

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Education

Stony Brook University - Ph.D. in Electrical and Computer Engineering — Jan. 2026 - Present

- Advisor : Prof. Petar Djurić (ECE), Prof. Sima Mofakham (Neurosurgery & ECE)

Pusan National University - M.S. in Information Convergence Engineering — Sep. 2020 - Aug. 2022

- Advisor : Prof. JinKyu Gahm
- Thesis : Developing a machine learning model to diagnose Atypical Parkinson's Syndrome through texture analysis of 3D brain magnetic resonance images

Pusan National University - B.S. in Computer Science and Engineering — Mar. 2014 - Aug. 2020

- Republic of Korea Army : Jul. 2015 - Apr. 2017

Experience

Stony Brook University - Research Assistant — Jan. 2026 - Present

- Mofakham-Mikell Lab
- Video understanding, multimodal LLMs, RL post-training, AI agent for clinical settings

Samsung Electronics - Research Engineer — Aug. 2022 - Aug. 2026

- Advanced Process Development Team, Semiconductor R&D Center
- Building foundation model for semiconductor nano-structure electron microscopy images

University of California, Santa Barbara - Graduate Researcher — Aug. 2025 - Dec. 2025

- Peng Li Lab
- Bayesian optimization with LLMs, Graph representation learning

Pusan National University - Research Assistant — Mar. 2020 - Aug. 2022

- Image Computing and Machine Learning Lab
- Topographic iron-distribution pattern analysis for atypical Parkinsonian syndrome diagnosis

Korea Electric Power Research Institute - Research Intern — Sep. 2019 - Feb. 2020

- Information Communication Technology Solution Lab
- Autonomous driving robot in underground electric power station for safety inspection

Publications

Under Review & Pre-prints

Confidence Inflation from RL Post-Training in Video Language Models — Sep. 2026

- **YunSoo Kim**, Jiacheng Qiu, Petar Djurić, Sima Mofakham
- *Under review, ICLR 2027*

Less Data, Better Timing: Student–Curriculum Coupling for VLM On-Policy Distillation in Temporal Video Grounding — Sep. 2026

- Jiacheng Qiu, **YunSoo Kim**, Ruichen Xu, Jian Luo, Petar Djurić, Sima Mofakham
- *Under review, ICLR 2027* (link)

ICU-TRACE: Forecasting of Agitation in the ICU Setting from Bedside Video Using a Temporal Risk-Aware Causal Encoder — Aug. 2026

- **YunSoo Kim**, Noah Lubin, Yindong Hua, Morsal Mosallami Aghili, et al.

Radiomic Features of Pre-embolization CTA and MRA Are Associated with Immediate Angiographic Occlusion of Brain Arteriovenous Malformations Aug. 2026

· Morsal Mosallami Aghili, Alfonse Mazzarella, Noah Lubin, **YunSoo Kim**, et al.

· *Under review, Journal of NeuroInterventional Surgery Advances*

Conference Proceedings

SMART-ICU: Monitoring and Predictive Modeling of Behavioral and Physiological Deterioration in the ICU Jul. 2026

· **YunSoo Kim**, Noah Lubin, Morsal Mosallami Aghili, Charles Mikell, Sima Mofakham

· *NIH Brain Behavior Quantification and Synchronization workshop 2026 (**NIH BBQS**)*

Forecasting Agitation in Severe Traumatic Brain Injury Through Temporal Facial Motion Dynamics May. 2026

· Noah Lubin, Yindong Hua, **Yunsoo Kim**, Morsal Mosallami Aghili, Charles Mikell, Sima Mofakham

· *American Society for Stereotactic and Functional Neurosurgery (**AASFN**)*

Differentiating Parkinsonian Syndromes Using Distinctive Brain Iron Accumulation Patterns in Susceptibility Weighted Image (SWI) Jan. 2022

· **YunSoo Kim**, JaeHyeok Lee, JinKyu Gahm

· *IEEE International Conference on Big Data and Smart Computing (**BigComp**) (link)*

Steel surface defect classification using ResNet50 Dec. 2021

· WonJune Choi, **YunSoo Kim**, JeongWon Jo, DongHyong Lee, et al.

· *Korea Software Congress 2021*

Surface-based Analysis of Subcortical structures in SWI for Atypical Parkinsonian Syndromes Jun. 2021

· **YunSoo Kim**, JaeHyeok Lee, JinKyu Gahm

· *Organization of Human Brain Mapping (**OHBM**)*

Journal Publications

Automated Differentiation of Atypical Parkinsonian Syndromes Using Brain Iron Patterns in Susceptibility Weighted Imaging Mar. 2022

· **YunSoo Kim**, JaeHyeok Lee, JinKyu Gahm

· *Diagnostics (link)*

Pose Classification and Correction System for At-home Workouts Sep. 2021

· JaeMin Kang, SeongSu Park, **YunSoo Kim**, JinKyu Gahm

· *Journal of the Korea Institute of Information and Communication Engineering (**JKIICE**)*

Multiple Sclerosis Lesion Detection using 3D Autoencoder in Brain Magnetic Resonance Images Aug. 2021

· WonJune Choi, SeongSu Park, **YunSoo Kim**, JinKyu Gahm

· *Journal of Korea Multimedia Society*

MRI Image Super Resolution through Filter Learning Based on Surrounding Gradient Information in 3D Space Feb. 2021

· SeongSu Park, **YunSoo Kim**, JinKyu Gahm

· *Journal of Korea Multimedia Society*

Patents

Method for Matching Semiconductor Sample Frontside and Backside Pair Image of Scanning Electron Microscopy for Automated Sample Preparation and Inspection 2025

- **YunSoo Kim**, YoonSung Bae, BeomSoo Park, SangHo Yoon, et al.
- KR 10-2025-0130253 (pending)

Method for Automatically Measuring Semiconductor Structure based on Learning of Semiconductor Image Segmentation Foundation Model 2024

- SangHo Yoon, SuBong Shon, **YunSoo Kim**, JoonYoung An, et al.
- KR 10-2024-0109944 (pending), US application (pending)

Electronic Device and Operation Method Thereof 2024

- MinSeok Kim, **YunSoo Kim**, SangBum Park, JoonHyeok Park, et al.
- KR 10-2024-0001688 (pending)

Method and Apparatus for Generating Basal Ganglia Mask Emphasizing Fe Component in Brain using T1 MRI and SWI, and Classification System of Atypical Parkinsonian Syndromes using the same 2022

- JaeHyeok Lee, JinKyu Gahm, **YunSoo Kim**
- KR 10-2662563 (registered, 2024)

System and Method for Classification of Steel Surface Defect Using Deep Learning Model 2021

- JinKyu Gahm, WonJune Choi, **YunSoo Kim**
- KR 10-2762749 (registered, 2025)

Program for Training and Validating a Deep Learning Model for Classifying Surface Defects in Cold-Rolled Steel Sheets 2021

- **YunSoo Kim**, JinKyu Gahm, WonJune Choi
- KR C-2021-060440 (registered, 2021, SW Copyright)

Awards

Samsung Best Paper Award - Grand Prize May. 2024

- TEM image segmentation modeling for automatically measuring core structure of semiconductor device
- CEO of Samsung Electronics

Chatbot Contest with Microsoft - Encouragement prize Jul. 2020

- Director of Korean Institute of Information Scientists and Engineers

Best Intern Award - 2nd prize Feb. 2020

- President of Korea Electric Power Research Institute